

An Internship Report

Topic

AI Deployment & Automation

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

**In partial fulfilment of the requirement of degree of
B.Tech CSE**



By

Roll Number of Student:2410030051

Name of Student: (AMAN SHARMA)

Batch: 2026-27

2026-27

CANDIDATE'S DECLARATION

I, **AMAN SHARMA** do hereby declare that the internship report titled “**AI Deployment & Automation**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: AMAN SHARMA

Roll No. 2410030051

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **AI Deployment & Automation**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature

Student Name AMAN SHARMA

Roll No..2410030051

TABLE OF CONTENTS

1. Candidate's Declaration.....	i
2. Acknowledgement.....	ii
3. Internship Completion Certificate.....	-
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organization Profile.....	2
4.3 Problem Statement.....	3
4.4 Project Objectives.....	3
4.5 Scope of the Project.....	4
4.6 Technologies and Tools Used.....	4
4.7 System Architecture.....	5
4.8 Methodology.....	6
4.9 Expected Outcomes.....	7
4.10 Certificates of Completion and Communication Proof.....	8
5. Bibliography/References.....	9

3. INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Aman Sharma

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

AI Deployment & Automation

During the internship, the learner demonstrated proficiency in:

- Foundations of AI Deployment & Production Readiness & Advanced API Deployment with FastAPI
- Containerization with Docker & Introduction to MLOps & CI/CD Pipelines
- Cloud Deployment Basics (AWS SageMaker, Google Vertex AI, Azure ML) & Deploying Generative AI Applications
- Building Advanced AI Agents with LangChain & LlamaIndex & AI-Powered Automation Workflows
- Integrating AI into Web/Mobile Apps & Security and Compliance in AI Deployment
- Container Orchestration with Kubernetes & Monitoring, Logging & Observability
- Performance Optimization & Cost Management & Model Drift Detection & Retraining & Advanced RAG & Agentic Workflows
- Case Studies & Best Practices in AI Deployment & Project: Deploy a Full-Stack RAG Application & Project: AI Agent for Business Automation & Capstone Project: End-to-End AI Solution for Business Problem

This certification recognizes the successful completion of the internship requirements.



Issued on: Jul 31, 2026
Certificate ID: 2026-E31C2B5F43



Mitu Swain
General Manager
(Learning & Development)
EduSkills

Internship Offer Letter



Ref No: C17Y2026M081522484

Internship Offer Letter

Student Name : Aman Sharma
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : AI Deployment & Automation
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU69e50fbaba77a1776619450

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	AI Deployment Foundations & FastAPI Model Serving	Introduction to AI lifecycle, production readiness, MLOps terminology, environment setup, and advanced API deployment using FastAPI including model integration and asynchronous operations.
Week 2	Containerization & CI/CD for MLOps	Docker architecture, Dockerfiles, networking, volumes, introduction to MLOps principles, and implementing CI/CD pipelines for machine learning workflows.
Week 3	Cloud AI Deployment & Generative AI Serving	Deployment on cloud platforms like Amazon SageMaker, Google Vertex AI, and Azure Machine Learning, along with deploying Generative AI applications, handling LLM challenges, vLLM serving, and model quantization.
Week 4	AI Agents & Automation Workflows	Building advanced AI agents using LangChain and LlamaIndex, agent orchestration with LangGraph, and implementing AI-powered hyper-automation workflows.
Week 5	AI Integration, Security & Compliance	Integrating AI into web and mobile apps using Streamlit and Gradio demos, securing AI systems, handling threats, ensuring privacy, and bias mitigation strategies.
Week 6	Kubernetes & Observability in Production AI	Container orchestration using Kubernetes, working with Pods, Deployments, Services, and implementing monitoring & logging with Prometheus and Grafana.
Week 7	Performance Optimization, Drift & Advanced RAG	Model performance tuning, cloud cost optimization, model drift detection using Evidently AI, retraining strategies, advanced RAG architectures, LlamaIndex agents, and vector database integrations.
Week 8	Applied AI Projects & Capstone Projects	Real-world case studies (e-commerce & healthcare), full-stack RAG application development (FastAPI backend + frontend), AI agent for business automation, and complete end-to-end production AI solution including model development, API deployment, monitoring, and scaling.
Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of “**AI Deployment & Automation**”. The internship was carried out from June 2026 to August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical understanding of Artificial Intelligence and Machine Learning and their real-world, production-grade deployment. Over the course of the programme, the learner was exposed to industry-relevant tools and practices used to take AI/ML models from a research or prototype stage to a fully deployed, monitored, and scalable production system.

The internship followed a structured, week-wise curriculum that combined conceptual learning with hands-on assignments, weekly assessments, project documentation, and a final capstone project. On successful completion of all the requirements, including the final assessment test, a Virtual Internship Certificate was awarded by EduSkills Academy.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. The organization collaborates with leading technology companies and academic institutions to design industry-aligned virtual internship curricula that help engineering students strengthen their employability and gain practical exposure to emerging technology domains such as Artificial Intelligence, Cloud Computing, Cybersecurity, and MLOps, among others.

4.3 Problem Statement

Although a large number of engineering students learn to build Machine Learning and Deep Learning models during their academic coursework, very few get exposure to the practices required to deploy such models into real, production-ready systems. Building an

accurate model is only one part of the AI lifecycle; making that model available as a reliable, secure, scalable, and monitorable service is an equally important and often overlooked skill. The problem addressed by this internship was to understand and practically implement the complete lifecycle of AI deployment – from serving a model through an API, containerizing it, deploying it on the cloud, orchestrating it at scale, integrating it into applications, and finally monitoring and optimizing it in production.

4.4 Project Objectives

- To understand the complete AI/ML deployment lifecycle and MLOps terminology used in the industry.
- To learn advanced API deployment of AI models using FastAPI, including asynchronous operations.
- To gain hands-on experience with Docker containerization and CI/CD pipelines for ML workflows.
- To deploy AI/Generative AI models on cloud platforms such as Amazon SageMaker, Google Vertex AI, and Azure Machine Learning.
- To build AI agents and automation workflows using LangChain, LlamaIndex, and LangGraph.
- To integrate AI models into web and mobile applications using Streamlit and Gradio, while ensuring security, privacy, and bias mitigation.
- To learn container orchestration with Kubernetes and implement observability using Prometheus and Grafana.
- To perform model performance optimization, drift detection, and build advanced Retrieval-Augmented Generation (RAG) systems.
- To apply all the above concepts in a real-world, end-to-end capstone project.

4.5 Scope of the Project

The scope of this internship spans the entire AI deployment pipeline – starting from serving a trained model through a REST API, packaging it into containers, deploying it to cloud infrastructure, orchestrating it using Kubernetes, and finally monitoring its performance in

production. It also covers the development of AI agents and Retrieval-Augmented Generation (RAG) based applications for real-world business use cases such as e-commerce and healthcare. The internship was limited to a virtual, simulation/lab-based environment provided by EduSkills Academy and did not involve deployment on any live organizational production system.

4.6 Technologies and Tools Used

- Programming & API Development: Python, FastAPI
- Containerization & Orchestration: Docker, Kubernetes
- Cloud Platforms: Amazon SageMaker, Google Vertex AI, Azure Machine Learning
- Generative AI & Agent Frameworks: LangChain, LlamaIndex, LangGraph, vLLM
- Application Integration: Streamlit, Gradio
- Monitoring & Observability: Prometheus, Grafana, Evidently AI
- CI/CD & MLOps: CI/CD pipelines for machine learning workflows
- Data & Retrieval: Vector Databases, Advanced RAG architectures

4.7 System Architecture

The general architecture followed during the internship for deploying an AI solution can be summarized in the following layers:

- Model Layer: A trained Machine Learning / Generative AI model (including LLMs) forms the core intelligence layer.
- API Layer: The model is exposed as a service using FastAPI, supporting synchronous and asynchronous requests.
- Containerization Layer: The API and its dependencies are packaged into a Docker container for portability and consistency across environments.

- **Orchestration Layer:** Kubernetes manages the deployment, scaling, and networking of containers (Pods, Deployments, Services) in a production-like cluster.
- **Cloud Layer:** The containerized application is deployed on cloud AI platforms such as Amazon SageMaker, Google Vertex AI, or Azure Machine Learning.
- **Application/Agent Layer:** AI agents built with LangChain, LlamaIndex and LangGraph, along with front-ends built using Streamlit/Gradio, consume the deployed model APIs.
- **Monitoring Layer:** Prometheus and Grafana are used for logging and monitoring, while Evidently AI is used for detecting model drift and triggering retraining.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, weekly assessments, and project submissions, as summarized in the table below.

Week	Module	Description
1	AI Deployment Foundations & FastAPI Model Serving	Introduction to AI lifecycle, production readiness, MLOps terminology, environment setup, and advanced API deployment using FastAPI including model integration and asynchronous operations.
2	Containerization & CI/CD for MLOps	Docker architecture, Dockerfiles, networking, volumes, introduction to MLOps principles, and implementing CI/CD pipelines for machine learning workflows.
3	Cloud AI Deployment & Generative AI Serving	Deployment on cloud platforms like Amazon SageMaker, Google Vertex AI, and Azure Machine Learning, along with deploying Generative AI applications, handling LLM challenges, vLLM serving, and model quantization.
4	AI Agents & Automation Workflows	Building advanced AI agents using LangChain and LlamaIndex, agent orchestration with LangGraph, and implementing AI-powered hyper-automation workflows.
5	AI Integration, Security & Compliance	Integrating AI into web and mobile apps using Streamlit and Gradio demos, securing AI systems, handling threats, ensuring privacy, and bias mitigation strategies.
6	Kubernetes & Observability in Production AI	Container orchestration using Kubernetes, working with Pods, Deployments, Services, and implementing monitoring & logging

		with Prometheus and Grafana.
7	Performance Optimization, Drift & Advanced RAG	Model performance tuning, cloud cost optimization, model drift detection using Evidently AI, retraining strategies, advanced RAG architectures, LlamaIndex agents, and vector database integrations.
8	Applied AI Projects & Capstone Projects	Real-world case studies (e-commerce & healthcare), full-stack RAG application development (FastAPI backend + frontend), AI agent for business automation, and complete end-to-end production AI solution including model development, API deployment, monitoring, and scaling.

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and assignments. In the final week, a Final Assessment Test was conducted, and the overall grade was based on performance across weekly assessments, project submissions, and the final test.

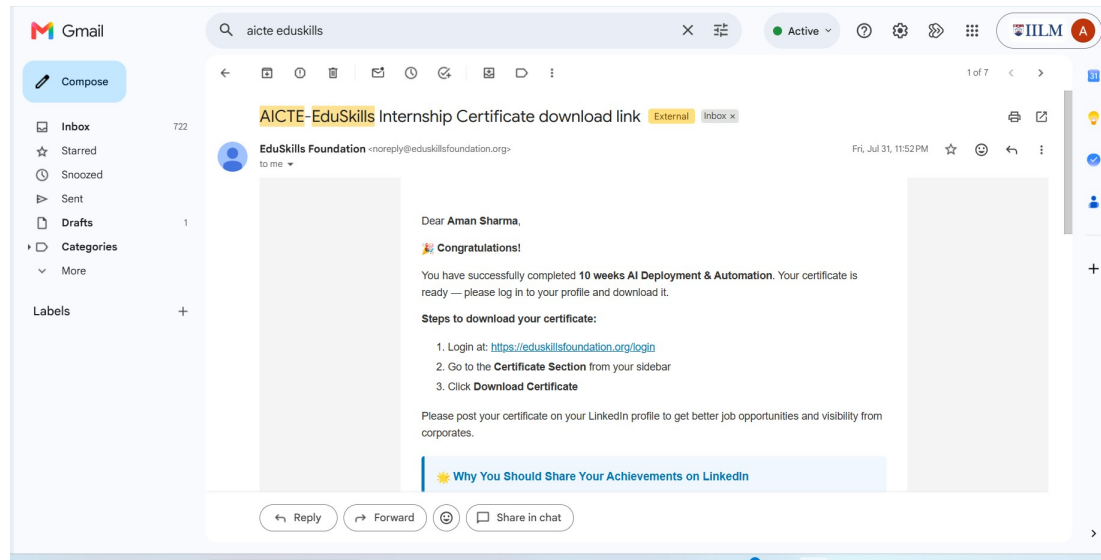
4.9 Expected Outcomes

- Ability to independently deploy an AI/ML model as a production-ready API using FastAPI.
- Practical understanding of containerization, CI/CD, and Kubernetes-based orchestration for ML workloads.
- Hands-on exposure to deploying models on major cloud AI platforms.
- Capability to design and build AI agents and automation workflows for business use cases.
- Understanding of AI security, privacy, and bias-mitigation practices while integrating AI into applications.
- Skill to monitor deployed models in production and detect/handle model drift.
- A completed, end-to-end capstone project demonstrating a full-stack RAG application and an AI agent for business automation.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 10-weeks AICTE –

EduSkills Virtual Internship Program in AI Deployment & Automation, are attached in Chapter 3 of this report.





शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Aman Sharma

IILM University, Greater Noida

has successfully completed the 8-weeks

AI Deployment & Automation Virtual Internship

During June - August 2026

Supported by



EduSkills
ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4d8488f5f48aa0e21bd6

Student ID: STU69e50fbaba77a1776619450



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, “AICTE – EduSkills Virtual Internship Program – AI Deployment & Automation,” Internship Offer Letter, Ref No. C17Y2026M081511743, 2026.
2. EduSkills Academy, “Certificate of Internship – AI Deployment & Automation,” Certificate ID: 2026-E31C2B5F43, Issued on Jul 31, 2026
3. FastAPI Documentation, <https://fastapi.tiangolo.com/>
4. Docker Documentation, <https://docs.docker.com/>
5. Kubernetes Documentation, <https://kubernetes.io/docs/>
6. LangChain Documentation, <https://python.langchain.com/>
7. AWS SageMaker, Google Vertex AI, and Microsoft Azure Machine Learning official documentation.